Quadriceps Harvest Site Closure is Associated with Increased Patellofemoral Contact Pressures: A Biomechanical Analysis

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ABSTRACT

Purpose: To evaluate the effect of quadriceps tendon (QT) autograft harvest and donor-site closure on patellofemoral contact pressures in a cadaveric model.

Methods: Eight fresh-frozen cadaveric hemipelvis-to-foot specimens were used. Thin-film pressure sensors were inserted into the patellofemoral joint via parapatellar arthrotomy. Baseline patellofemoral mean contact pressure, peak contact pressure, and contact area were recorded at 0°, 30°, 60°, 90°, and 110° of knee flexion. Measurements were repeated after harvesting a QT, and again after loose closure and tight closure of the donor-site defect, with closure order randomized. Paired comparisons were performed.

Results: Closure of the harvest site was associated with significantly elevated pressures at higher flexion angles. Tight closure increased peak pressure at 60° (median 42.0 vs 29.5, PSI; p=0.012) and 110° (82.0 vs 47.2 PSI; p=0.012), and mean pressure at 90° (38.5 vs 24.9 PSI; p=0.025) and 110° (35.5 vs 30.1 PSI; p=0.012). Loose closure increased peak pressure at 60° (median 36.0 vs 29.5 PSI; p=0.025) and 110° (66.2 vs 47.2 PSI; p=0.012), and mean pressure at 90° (33.1 vs 24.9 PSI; p=0.050). Contact area did not differ significantly across any condition or flexion angle.

Conclusion: QT autograft harvest alone did not increase patellofemoral contact pressures. However, both loose and tight closure of the donor-site defect significantly elevated mean and peak pressures at higher flexion angles (60°–110°), suggesting that closure may be an underrecognized contributor to postoperative patellofemoral symptoms and chondral damage following QT harvest.

Level of Evidence: Level V, cadaveric biomechanical study.

INTRODUCTION

Anterior cruciate ligament (ACL) rupture is among the most common knee injuries encountered in active individuals and athletes, with an estimated annual incidence of approximately 1 in 3,000 individuals in the United States, resulting in more than 150,000 new injuries each year[1]. ACL reconstruction has become one of the most frequently performed procedures in orthopaedic sports medicine[1,2]. As surgical techniques have evolved, increasing emphasis has been placed on restoring native ACL anatomy and biomechanics while minimizing graft-related morbidity and optimizing postoperative outcomes[2,3].

Autograft options for ACL reconstruction most commonly include bone-patellar tendon-bone (BPTB), hamstring tendon, and quadriceps tendon (QT) grafts. Although BPTB and hamstring autografts have historically been the predominant graft choices, interest in QT autografts has increased substantially over the past decade owing to favorable biomechanical characteristics, greater graft cross-sectional area, preservation of hamstring function, and clinical outcomes comparable to traditional graft options[4,5]. Data from an international summit of ACL surgeons demonstrated that QT autografts accounted for approximately 10% of graft utilization by 2011, reflecting a marked increase from earlier reports and highlighting the growing acceptance of this graft option worldwide[2,5]. Furthermore, QT autografts have become particularly attractive in pediatric and revision settings because they provide a robust soft-tissue graft while avoiding some of the limitations associated with alternative autografts[4,5].

Despite these advantages, donor-site morbidity following QT harvest remains an important consideration. A recent systematic review including more than 2,300 patients undergoing ACL reconstruction with QT autografts reported an overall complication rate exceeding 10%, with persistent anterior knee pain representing the most common postoperative complaint[4]. While the etiology of postoperative anterior knee pain following QT harvest remains

incompletely understood, alterations in patellofemoral joint biomechanics including increased joint contact pressure represent a plausible contributing mechanism[6,7]. We have also observed anecdotal clinical evidence supporting the potential risk of QT in a 15-year-old male who underwent ACL reconstruction with a QT autograft and harvest site closure. Arthroscopic evaluation at the index procedure demonstrated intact patellofemoral cartilage; however, during postoperative rehabilitation he developed progressively worsening anterior knee pain while performing passive stretching exercises. Repeat arthroscopic evaluation subsequently revealed extensive patellofemoral chondromalacia (Figure 1). Although anecdotal, many research studies link mechanical overload with cartilage degeneration, therefore, this case further underscores the need to better understand the biomechanical consequences of QT harvest on the patellofemoral joint.

While previous cadaveric studies have found that BPTB graft harvest, with or without closure of the donor-site defect, did not significantly alter patellofemoral contact pressures or contact area immediately following graft procurement[9,10], there is no biomechanical studies evaluating patellofemoral contact pressures after QT harvest. Therefore, the purpose of this cadaveric biomechanical study was to evaluate patellofemoral contact pressures before and after QT harvest and to determine the effect of donor-site closure on patellofemoral joint mechanics. We hypothesized that closure of the QT harvest donor-site would increase patellofemoral contact pressures compared with the native state.

METHODS

Cadaveric Specimen Preparation

Ten fresh-frozen human cadaveric hemipelvis-to-foot specimens without prior injury or surgical history (mean age, sex, etc) were thawed at 4° C for 48 hours prior to testing. Hemipelvis specimens were selected rather than isolated knees to preserve the native origins and insertions of all musculotendinous structures spanning the knee, including the quadriceps proximally and gastrocnemius distally. The specimens were securely mounted on testing stations using femoral transosseous pins that were attached to engineered posts. The pins were inserted lateral to medial while staying posterior to the vastus muscle and well proximal to the knee, to avoid interfering with the extensor mechanism tension. The tibia and ankle were left free to allow full flexion and extension of the knee (Figure 2). The knees were pre-conditioned using 100 cycles of full flexion and extension to eliminate tissue creep.

A small parapatellar lateral arthrotomy was performed and a thin-film pressure sensor (I-Scan 4000; Tekscan, Inc., South Boston, MA) was inserted between the patella and femur. Each pressure sensor was calibrated using I-Scan software and a two-point calibration method. The sensors operate on a 0-1500psi range, so calibration allowed for precision, while encompassing the expected forces. Sensors were loaded on an Instron with low and high calibration point forces of 22.24 N (5lbf) and 444.8 N (100lbf), respectively. Software sensitivity was manually adjusted based on readings at the high calibration point force.

The sensor was secured to the patella using 4-0 monofilament suture at the corners of the patella, such that the sensor glided across the femur with the patella during flexion and extension of the knee. The parapatellar arthrotomy was closed around the sensor handle using interrupted

No. 0 braided figure-of-eight sutures to restore tension across the patellofemoral joint. Attention was paid to approximate the tissues back to their baseline tension with no excessive overlap to avoid over tensioning the capsule.

Following arthrotomy, the patellofemoral cartilage was examined to ensure no or minimal arthritis. Two of the 10 specimens had end-stage patellofemoral arthritis and severe osteophyte formation. To minimize bias, these two specimens were excluded from the study, leaving eight specimens that were utilized.

Patellofemoral Contact Pressure Measurement

Each knee was passively cycled through full range of motion to ensure smooth gliding of the sensor prior to measurement. Starting with the knee at full extension, baseline patellofemoral contact pressure and area, were measured at 0°, 30°, 60°, 90°, 110°. A goniometer was used to determine flexion angles. The knees were loaded with passive flexion only by an examiner and suspended at each desired flexion angle to allow measurement. To permit physiologic tibial rotation during knee motion, the distal leg was suspended in a sling around the ankle, allowing unrestricted axial rotation throughout flexion and extension. Each measurement was held for 5 seconds (collecting 100 measurements per second), and the average of each parameter for the 5 seconds was used (500 measurements per flexion angle).

Quadriceps Graft Harvest and Closure

After collection of baseline parameters, the quadriceps tendon was harvested per routine surgical technique. A centrally based full-thickness quadriceps tendon graft measuring 10 mm in width and 70 mm in length was harvested through a longitudinal incision using standard instrumentation. Post-harvest biomechanical testing was repeated using the same data acquisition methods.

The resulting donor-site defect was then closed with interrupted No. 0 braided figure-of-eight sutures in a standard fashion to reapproximate the remaining tendon edges. We defined loose closure as a figure of eight stitch at every 10 mm interval, and tight closure as a figure of eight stitch every 5 mm interval. Suture bites were taken in full thickness at approximately 2-4 mm from the harvest edges. All biomechanical testing was repeated again after loose closure and tight closure.

Sequence of Testing

All samples had baseline testing prior to harvest collection, followed by testing after harvest. With regards to loose and tight closure, the order was randomized to minimize bias; four samples had loose followed by tight closure testing, and four samples had tight followed by loose closure testing.

Statistical Analysis

An a-priori power analysis was performed for the primary outcome of average patellofemoral contact pressure measured before graft harvest and after donor-site closure. Based

on pilot testing in a single cadaver specimen, a mean increase in contact pressure of approximately 20 PSI was observed. To provide a conservative estimate, the standard deviation of the paired differences was assumed to be 20 PSI, corresponding to an anticipated standardized effect size (Cohen's d) of 1.0. Using a two-sided α of 0.05 and a desired power of 80%, a minimum sample size of eight paired specimens was required.

The primary outcomes were mean and peak patellofemoral contact pressures. Contact area was evaluated as secondary outcomes. Continuous variables are presented as median and interquartile range (IQR). Because each specimen served as its own control and the data were not assumed to be normally distributed, paired comparisons between pre-harvest and post-closure measurements were performed using the Wilcoxon signed-rank test. All statistical tests were two-sided, and statistical significance was defined as $p < 0.05$. Statistical analyses were performed using Stata 15.1 (StataCorp LLC, College Station, TX, USA).

Ethical Approval

This study utilized de-identified fresh-frozen human cadaveric specimens obtained from a licensed tissue bank. Research involving deceased individuals does not constitute human subjects research under 45 CFR 46.102(e)(1), and institutional review board approval was therefore not required. For the clinical case presented in Figure 1, written informed consent for the use and publication of arthroscopic images and de-identified clinical information was obtained from the patient's legal guardian, with assent from the patient.

RESULTS

Specimen Characteristics

The available donor pool consisted of ten hemipelvis-to-foot specimens from six male and four female donors with a mean age of 72.1 years (range, 60-81 years). Eight specimens underwent biomechanical testing as two specimens had severe patellofemoral osteoarthritis on gross evaluation and were excluded.

Effect of Quadriceps Tendon Harvest on Patellofemoral Contact Mechanics

Compared with baseline, quadriceps tendon harvest did not significantly alter patellofemoral **mean contact pressure** at 60° (19.3 vs. 17.4 PSI, $p = 0.6744$) or 90° of knee flexion (24.9 vs. 22.7 PSI, $p = 0.1609$) (Table 1). However, at 110° of flexion, mean contact pressure was significantly reduced following graft harvest (30.1 vs. 28.2 PSI, $p = 0.0172$).

Similarly, peak contact pressure was unchanged following harvest at 60° of flexion (29.5 vs. 28.0 PSI, $p = 0.4002$), but was significantly lower at 90° (40.5 vs. 36.0 PSI, $p = 0.0140$) (Table 2). Peak contact pressure at 110° also demonstrated a reduction after harvest (47.2 vs. 46.8 PSI), although this difference did not reach statistical significance ($p = 0.1235$, figure 3).

Effect of Donor-site Closure on Patellofemoral Contact Mechanics

Loose Donor-site Closure

Loose donor-site closure significantly increased mean patellofemoral contact pressure at 90° compared with baseline (33.1 [25.0, 48.8] vs. 24.9 [22.7, 39.5] PSI; p = 0.0497), with no significant differences observed at the remaining flexion angles (Table 1). Loose closure also significantly increased peak patellofemoral contact pressure at 60° (36.0 [25.7, 56.3] vs. 29.5

[20.5, 44.5] PSI; p = 0.025) and 110° (66.2 [52.0, 102.0] vs. 47.2 [45.0, 88.0] PSI; p = 0.012), with no significant differences at the remaining flexion angles (Table 2, Figure 3).

Tight Donor-site Closure

Tight donor-site closure significantly increased mean patellofemoral contact pressure at 90° (38.5 [29.5, 48.5] vs. 24.9 [22.7, 39.5] PSI; p = 0.025) and 110° (35.5 [33.0, 72.2] vs. 30.1 [27.5, 57.0] PSI; p = 0.012) compared with baseline, with no significant differences observed at the remaining flexion angles (Table 1). Tight closure similarly increased peak patellofemoral contact pressure at 60° (42.0 [28.0, 55.0] vs. 29.5 [20.5, 44.5] PSI; p = 0.012) and 110° (82.0 [55.5, 120.7] vs. 47.2 [45.0, 88.0] PSI; p = 0.012), whereas no significant differences were identified at the remaining flexion angles (Table 1, Figure 3).

Contact Area

Median patellofemoral contact area did not differ appreciably following quadriceps tendon harvest or after either loose or tight donor-site closure at any knee flexion angle (Table 3).

DISCUSSION

The primary finding of this cadaveric biomechanical study was that quadriceps tendon autograft harvest alone did not significantly increase patellofemoral contact pressures. Closure of the harvest defect, however, was associated with increased mean and peak patellofemoral contact pressures at higher degrees of knee flexion. Loose and tight closure increased patellofemoral loading relative to the pre-harvest state, while contact area did not differ significantly across

testing conditions. These results suggest that donor-site closure, rather than harvest itself, may be the critical variable altering patellofemoral joint mechanics following quadriceps tendon autograft ACL reconstruction.

Quadriceps tendon autograft use has grown substantially in ACL reconstruction due to its favorable graft dimensions, reliable biomechanical properties, preservation of hamstring function, and improved clinical outcomes comparable to traditional autograft options [36096362, 41774566, 34494906]. Despite these advantages, donor-site morbidity continues to be a clinically meaningful concern. One review of more than 2,300 patients undergoing ACL reconstruction with quadriceps tendon autograft identified anterior knee pain as the most common postoperative complication [38145223]. This pain may be due to a combination of disruption of extensor mechanism integrity, subsequent quadriceps weakness, and articular cartilage degeneration [28888527, 25940814, 41774566]. The present findings support this framework by demonstrating that side-to-side closure of the quadriceps tendon donor site increases patellofemoral contact pressures, particularly in deeper flexion.

The flexion-angle dependence of these findings carries direct clinical relevance. Increases in mean and peak pressure were most pronounced at higher degrees of knee flexion, which is where patellofemoral joint reaction forces are greatest under normal physiologic circumstances **(Figure 3)** Patients report symptoms during squatting, kneeling, stair descent, and progressive rehabilitation, all activities that demand substantial knee flexion [35115309, 22267190]. Importantly, measurements in the present study were obtained during passive knee flexion rather than active quadriceps contraction. Active loading during functional tasks would be expected to generate considerably higher patellofemoral joint reaction forces [37272685, 22267190]. The pressure elevations observed in our passive model may therefore represent a conservative

estimate of the mechanical changes that occur clinically after donor-site closure. This flexion dependence is consistent with the clinical course of the patient described above. His anterior knee pain developed during passive stretching exercises rather than during resisted quadriceps activity, a loading condition closely approximating the passive flexion protocol used here. That symptoms arose under this comparatively low-demand condition, and that repeat arthroscopy demonstrated extensive patellofemoral chondromalacia in a joint with normal cartilage at the index procedure, raises the possibility that the pressure elevations measured after harvest-site closure are sufficient to produce chondral injury even without active quadriceps loading. This interpretation is supported by a substantial body of biomechanical literature demonstrating that supraphysiologic compressive loading induces chondrocyte death, proteoglycan and collagen matrix disruption, and surface fissuring, changes that progress over time toward the degenerative cartilage failure observed clinically in this patient [11032407, 22289896, 11562121, 16320827]. The precise magnitude and duration of loading required to produce these changes in patellar cartilage specifically remain undefined, however, as most foundational work in this area has employed femoral condyle or animal cartilage models rather than human patellar tissue [11032407, 22289896, 11562121]. A single case cannot establish causation, and unmeasured factors may have contributed. It does, however, illustrate the pattern our data would predict.

Our testing model was designed to approximate physiologic passive knee motion as closely as possible within a cadaveric setting. Hemipelvis-to-foot specimens rather than isolated knee preparations preserved the native musculotendinous attachments that span the joint. The distal limb was suspended in a sling to permit unrestricted tibial rotation during flexion rather than imposing a rigid tibial constraint. Patellofemoral contact mechanics are sensitive to extensor mechanism tension, soft-tissue restraint, and tibiofemoral rotation [11415841, 16951122,

29579721]. The preserved soft-tissue architecture and unconstrained distal limb strengthen the biomechanical relevance of the observed pressure changes.

The study's findings contrast with prior cadaveric investigations of bone-patellar tendon-bone autograft harvest. Central-third patellar tendon harvest, with or without donor-site closure, has not been shown to significantly alter patellofemoral contact pressures or contact area [1456369, 11799008, 9084346]. Mechanical consequences of quadriceps tendon harvest should not be assumed to mirror those of patellar tendon harvest. The quadriceps tendon serves as the proximal component of the extensor mechanism [37399632, 25940814]. Disruption or overconstrained repair of the tendon may create tensile forces across the patellofemoral articulation in a way that patellar tendon harvest does not. Differences in the shapes of these tendons and their spatial relationships with the patellofemoral joint could account for these differences.

This study has several limitations. The cadaveric design with a limited sample size prevents direct extrapolation to clinical outcomes. Testing was conducted under passive conditions and did not replicate active quadriceps contraction, hamstring co-contraction or gait mechanics. Only one graft size and two closure constructs were assessed, and partial-thickness harvests, varied graft dimensions, alternative suture configurations, or differences in closure tension could produce different biomechanical results. Acute cadaveric testing cannot account for the effects of postoperative healing, scar formation, tendon remodeling, or neuromuscular adaptation over time. Moreover, the average age size of our sample is higher than the average age of ACL reconstruction candidates. Finally, this study does not establish a causal relationship between donor-site closure and the development of anterior knee pain.

Quadriceps tendon autograft harvest alone did not increase patellofemoral contact pressures in this cadaveric model. However, donor-site closure significantly elevated mean and peak pressures at higher degrees of knee flexion. These results suggest that routine closure of the harvest defect may increase patellofemoral loading in a clinically relevant range of motion. Surgeons should consider the potential biomechanical consequences of closing the quadriceps tendon donor site, and future prospective clinical studies are needed to determine whether leaving the defect open reduces anterior knee pain or patellofemoral symptoms following ACL reconstruction with quadriceps tendon autograft.

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TABLES/FIGURES

Table 1. Patellofemoral mean contact pressure (PSI) across baseline, no-closure, and donor-site closure conditions

Flexion	Baseline	No Closure	p	Loose closure	p	Tight closure	p
0°	0.0 (0.0, 5.5)	0.0 (0.0, 6.5)	-	0.0 (0.0, 7.3)	-	0.0 (0.0, 7.4)	-
30°	13.5 (6.1, 15.0)	11.1 (4.5, 14.4)	0.5470	16.7 (12.6, 18.1)	0.3213	14.1 (11.4, 17.6)	0.2596
60°	19.3 (14.1, 27.5)	17.4 (14.1, 29.2)	0.6744	21.2 (17.3, 30.1)	0.3264	22.2 (18.5, 33.2)	0.1614
90°	24.9 (22.7, 39.5)	22.7 (18.7, 28.0)	0.1609	33.1 (25.0, 48.8)	0.0497	38.5 (29.5, 48.5)	**0.0251**
110°	30.1 (27.5, 57.0)	28.2 (21.7, 41.5)	**0.0172**	31.0 (30.0, 62.7)	0.6744	35.5 (33.0, 72.2)	**0.0116**

Values are presented as median (interquartile range). P values represent paired comparisons between each experimental condition and the baseline (pre-harvest) state using the Wilcoxon signed-rank test. Statistically significant differences ($p < 0.05$) are shown in bold.

Table 2. Patellofemoral peak contact pressure (PSI) at baseline, after quadriceps tendon harvest, and following loose and tight donor-site closure

Flexion	Baseline	Post Harvest	p	Loose closure	p	Tight closure	p
0°	0.0 (0.0, 6.2)	2.4 (0.0, 15.9)	0.2500	0.0 (0.0, 8.2)	0.6253	0.0 (0.0, 5.0)	0.2500
30°	18.5 (7.3, 23.0)	15.9 (7.5, 21.5)	0.3513	22.5 (17.9, 26.0)	0.5459	21.0 (17.4, 31.0)	0.0923
60°	29.5 (20.5, 44.5)	28.0 (16.5, 46.9)	0.4002	36.0 (25.7, 56.3)	**0.0251**	42.0 (28.0, 55.0)	**0.0115**
90°	40.5 (33.0, 80.0)	36.0 (27.5, 59.7)	**0.0140**	67.9 (42.0, 86.5)	0.0584	67.6 (49.0, 91.2)	0.0929
110°	47.2 (45.0, 88.0)	46.8 (38.6, 63.5)	0.1235	66.2 (52.0, 102.0)	**0.0116**	82.0 (55.5, 120.7)	**0.0117**

Values are presented as median (interquartile range). P values represent paired comparisons between each experimental condition and the baseline (pre-harvest) state using the Wilcoxon signed-rank test. Statistically significant differences ($p < 0.05$) are shown in bold.

Table 3. Patellofemoral contact area (pixels) at baseline, after quadriceps tendon harvest, and following loose and tight donor-site closure

Flexion	Baseline	Post Harvest	Loose closure	Tight closure
0°	0 (0, 0)	0 (0, 3)	0 (0, 3)	0 (0, 5)
30°	16 (2, 31)	17 (0, 21)	23 (11, 25)	22 (18, 27)
60°	53 (42, 54)	29 (23, 47)	51 (23, 56)	42 (37, 48)
90°	79 (69, 82)	60 (50, 85)	62 (50, 89)	56 (44, 87)
110°	73 (60, 97)	63 (61, 87)	65 (61, 103)	65 (49, 98)

Values are presented as median (interquartile range) pixel contact area at each knee flexion angle. One pixel corresponds to 1.61 mm^2. No significant differences in contact area were observed across conditions.

Figure 1. Arthroscopic images of 15-year-old male who underwent ACL reconstruction with a QT autograft with closure of the harvest site. Arthroscopic evaluation at the (A) index procedure demonstrated intact patellofemoral cartilage. (B, C) Repeat arthroscopic evaluation subsequently revealed extensive patellofemoral chondromalacia.

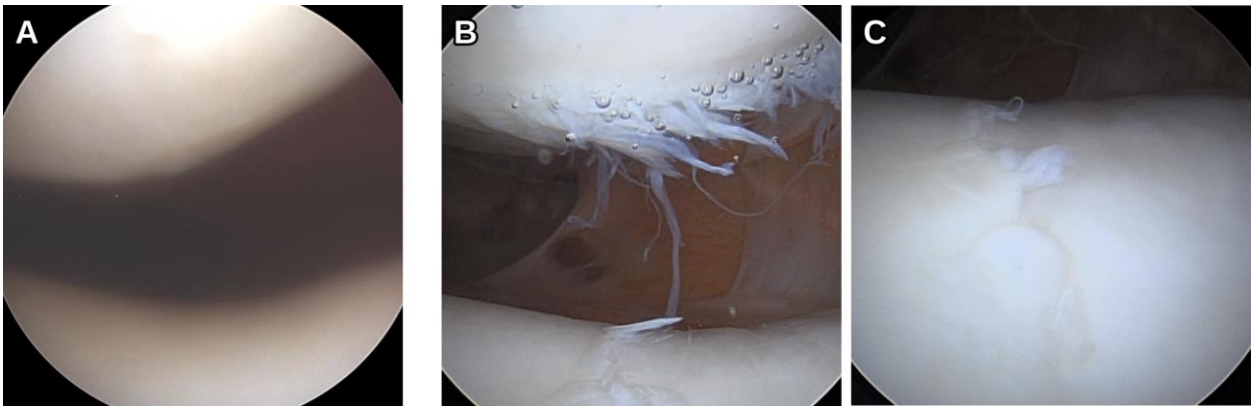


Figure 2. Photographs of the testing setup demonstrating A. Mounting of the specimens, B. Tight closure of the quadriceps harvest site, C. Closure of the arthrotomy site with the sensor handle exiting the arthrotomy.

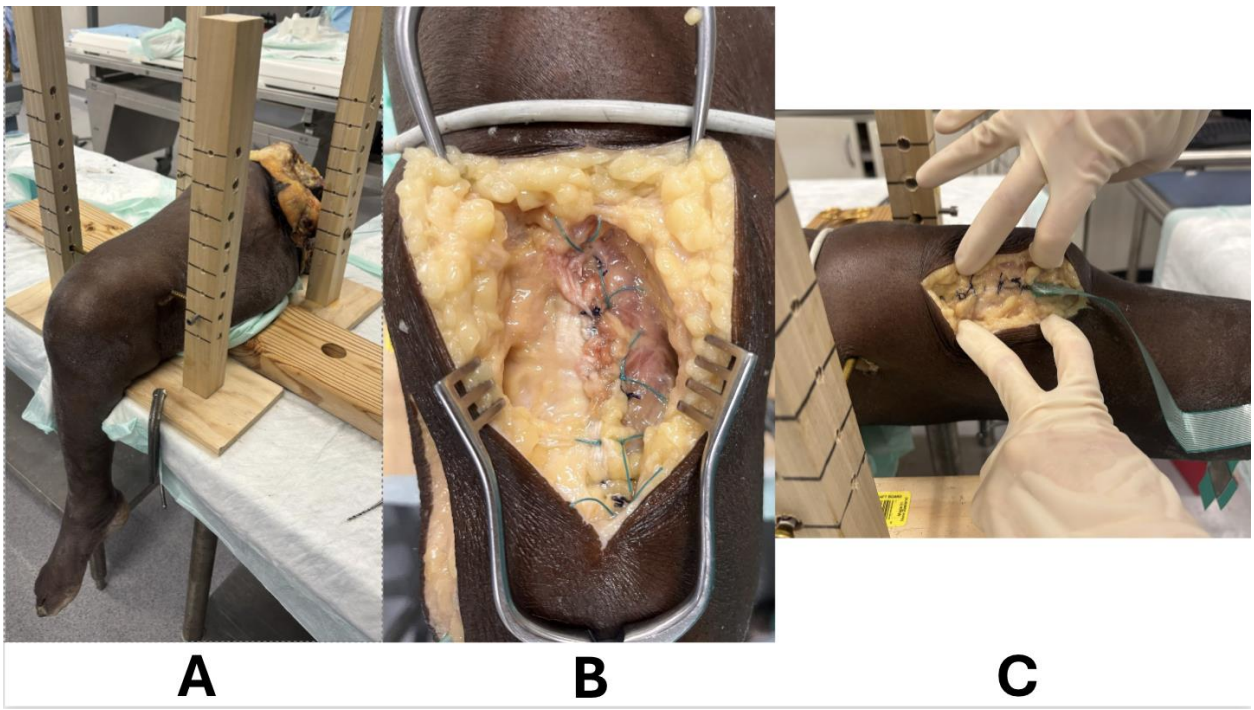


Figure 3. Contact pressure heat map of a single specimen demonstrating increasing patellofemoral contact pressure values after loose and tight closure compared to pre-harvest and no closure testing. The effect is most prominent at 90° and 110° of flexion.

